

Lecture Scribe — Transformation of Random Variables

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: 11 – Transformation of Random Variables

Source: Lecture Slides by Dhaval Patel, PhD

1. Learning Objectives (From Outline Slide)

The lecture covers:

1. Transformation of random variables.
2. Function of two random variables.
3. Illustrative example: derivation for the case $Z = X + Y$.

The key objective stated in the slides is: How to find the PDF of a new transformed random variable.

2. Definitions and Notation

Random Variables and Transformations

Let X be a random variable with known probability density function (PDF) $f_X(x)$. A transformed random variable is defined as:

$$Y = g(X) \tag{1}$$

The cumulative distribution function (CDF) of Y is:

$$F_Y(y) = P(Y \leq y) \tag{2}$$

If X has CDF $F_X(x)$, then transformation proceeds through the CDF relation shown in the slides.

Function of Two Random Variables

Given two random variables X and Y , a new variable may be defined as:

$$Z = g(X, Y) \tag{3}$$

Example considered in lecture: $Z = X + Y$.

3. Assumptions and Conditions

From the slides:

1. The PDF of the original random variable is known a priori.
2. The function $Y = g(X)$ is considered under:
 - Monotonically increasing case.
 - Monotonically decreasing case.
3. For the two-variable example: X and Y may be independent (as stated in problem statements).
4. Limits of transformation must be changed according to the transformation.

4. Transformation of One Random Variable

Problem Setup

Given: $Y = g(X)$. Goal: Find $f_Y(y)$.

Step 1 — Express CDF of Transformed Variable

From the slide derivation:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) \quad (4)$$

For the monotonic increasing case:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y)) \quad (5)$$

Step 2 — Differentiate to Obtain PDF

Differentiate with respect to y :

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y)) = f_X(g^{-1}(y)) \frac{dg^{-1}(y)}{dy} \quad (6)$$

Equivalently: $f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$ evaluated at $x = g^{-1}(y)$.

5. Worked Example — Transformation of One RV

Given: $X \sim \text{Uniform}(-1, 1)$, so $f_X(x) = \frac{1}{2}$ for $-1 < x < 1$.

Transformation: $Y = \sin(\frac{\pi}{2}X)$.

Step 1 — Inverse Transformation: $x = \frac{2}{\pi} \sin^{-1}(y)$.

Step 2 — Compute Derivative: $\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$.

Step 3 — Apply Formula:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi \sqrt{1-y^2}} = \frac{1}{\pi \sqrt{1-y^2}} \quad \text{for } -1 < y < 1. \quad (7)$$

6. Function of Two Random Variables

Problem Statement: Let $Z = X + Y$. Find the PDF of Z .

The slides specify:

1. Find $F_Z(z)$.
2. Case when X and Y are independent.
3. Example cases include: $X, Y \sim N(0, 1)$ or exponential distributions with parameter λ .

7. Derivation for $Z = X + Y$

Step 1 — Start from Definition: $F_Z(z) = P(X + Y \leq z)$.

Step 2 — Express Using Joint PDF:

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x, y) dx dy \quad (8)$$

Step 3 — Integration Representation:

$$\int \left(\int f_{X,Y}(x, y) dx \right) dy \quad \text{or} \quad \int \left(\int f_{X,Y}(x, y) dy \right) dx \quad (9)$$

Step 4 — Obtain PDF: $f_Z(z) = \frac{d}{dz} F_Z(z)$.

8. Summary of Results (From Slides)

1. **Transformation method:** Use CDF $\rightarrow$ Apply inverse transformation $\rightarrow$ Differentiate to obtain PDF.
2. **Final transformation formula:** $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$.
3. **For two random variables:** Express CDF using joint distribution, integrate over the region defined by transformation, and differentiate to obtain PDF.